# False Alarms from Kernel Tests on Time Series with Long Memory

## Where the tests fail, why they fail, and what to do about it

Call with Gareth Peters, Wednesday 4 November 2026

# Research problem and questions

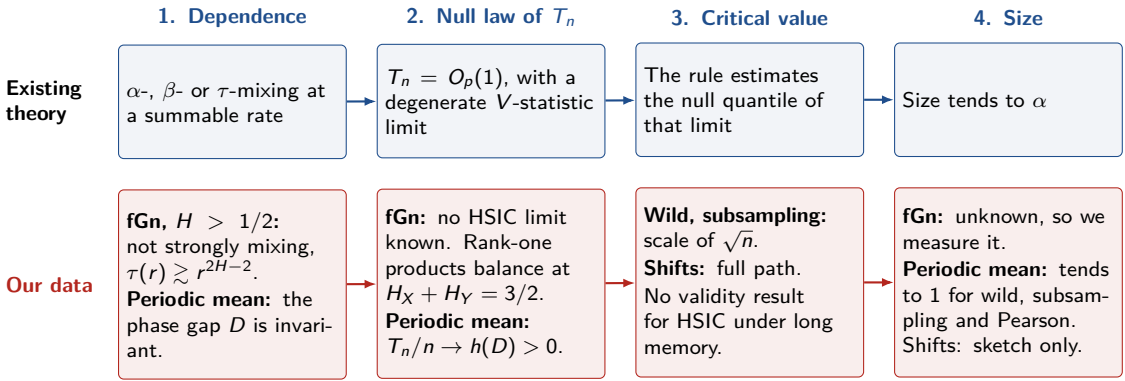
## Setting

We observe one path $(X_t, Y_t)_{t \le n}$ and test $H_0$: the processes $X$ and $Y$ are independent. Each process may have long memory, $\rho(k) \sim ck^{2H-2}$ with $H \in (1/2, 1)$. Our baseline is fractional Gaussian noise (fGn). The statistic is $T_n = nV_n$, the Gaussian-kernel HSIC $V$-statistic with median bandwidth. For split MMD, $H_0$ is that the two halves of one stationary series share a distribution. A rule estimates the critical value $\hat{c}_{n,1-\alpha}$ from the same path. The rule is calibrated if $\Pr_{H_0}(T_n > \hat{c}_{n,1-\alpha}) \to \alpha$.

## Research questions

1. **Scope of theory.** Do the published assumptions exclude these processes?
2. **Calibration.** For which statistic, critical-value rule, $H$ and $n$ does the size depart from $\alpha = 5\%$?
3. **Attribution.** Does the departure come from memory, from other process features (heavy tails, nonlinear dynamics, seasonality), or from their interaction?
4. **Procedure.** Which procedure keeps size and power, and what limits it?

# Why the 5% false-alarm guarantee is in doubt: its proof breaks at the first step




**Why** Evidence: source audit and Proposition 1

The top row is a chain: fast-fading dependence gives a limit for $T_n$, the limit sets the critical value, and that sets the size. fGn with $H > 1/2$ fails at the first box. The later steps may still hold, but nobody has proved them, and the null limit of HSIC under fGn is unknown. So we measure the size by simulation.

*$\to$ So the size must be measured. First, the objects we compare.*

# Three kinds of statistic, each needing a critical value

| Statistic | Input | Null hypothesis |
|---|---|---|
| **Split MMD** | One series | The two halves have the same distribution |
| **HSIC, distance covariance** | Two aligned series | The two processes are independent |
| **Pearson correlation** | Two aligned series | No linear link. This is the linear reference. |

**The critical-value rule is the part that uses the dependence**

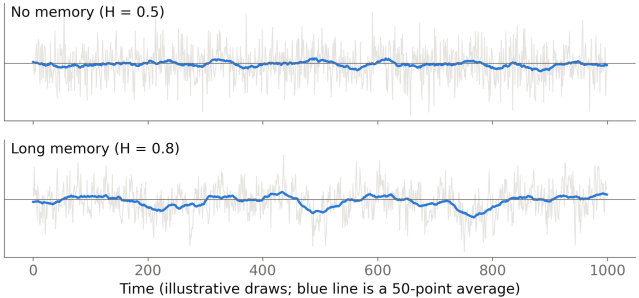
A rule must estimate the null quantile of the statistic from one dependent path. We therefore treat the rule as a factor in its own right, alongside the statistic.

**Why** Evidence: logic of the design

We run several rules on the same statistic. An oracle that size-adjusts the bare statistic then shows what each rule costs.

→ *The rules differ in how much of the path they use, which matters if memory is slow.*

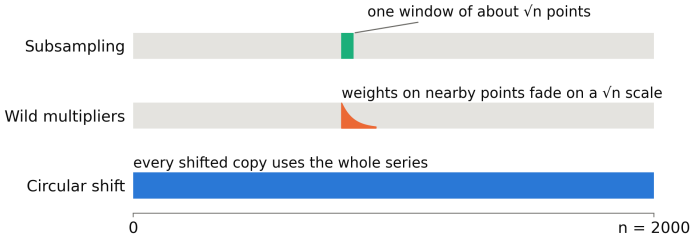
# Long memory: non-summable autocovariance reduces the information in a path




**Why** Evidence: exact calculation (Proposition 2)

The gap between two half-sample means has variance $(4 - 2^{2H})\, m^{2H-2}$. At $H = 0.5$ this shrinks like $1/m$. At $H = 0.8$ it shrinks only like $m^{-0.4}$. The paths are single simulated draws.

# Three critical-value rules: what each one uses from the path

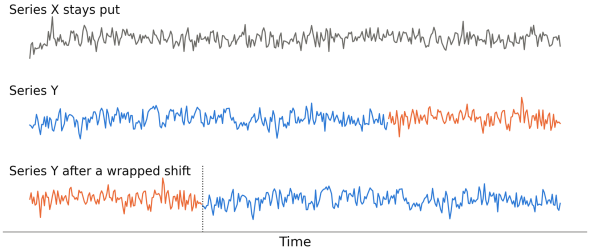



**Why** Evidence: explanation, consistent with the measured rates

- Subsampling uses windows of $\sqrt{n}$ points, which is 44 points at $n = 2000$. The wild rule weights lags on the same scale.
- Long memory moves slowly, so a short stretch can miss most of the movement, and the critical value can come out too low.

→ *This suggests a rule that uses the full path.*

# The circular shift breaks the cross-dependence and keeps each series' memory




**Why** Evidence: construction (Chwialkowski and Gretton, 2014)

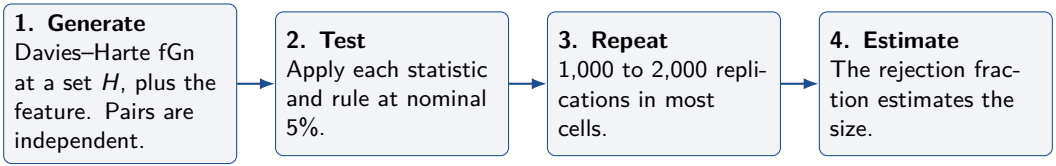
- The order inside $Y$ barely changes, so $Y$ keeps its own memory. Only the alignment with $X$ moves.
- Repeating this over many offsets gives the critical value from the full series. We exclude offsets below $0.1n$.
- Their result assumes mixing, and fGn with $H > 1/2$ is not mixing.

→ *Next: how we test these rules, and why this design.*

# How we test each question, and why this design

| **Question** | How we test it | Why this design |
|---|---|---|
| **1. Theory** | Read the stated conditions in the primary sources. Check fGn against each one. | A warning by implication appears only in the conditions, so we check them directly. |
| **2. Calibration** | Monte Carlo under a known null. Cross rule, $H$ and $n$. Compare Hermite ranks one and two. | No HSIC null limit is available, so we measure size. The rank calculation predicts where the scale changes. |
| **3. Attribution** | Factorial design: one process feature at a time, crossed with $H$. Holm correction. | It separates a feature's effect alone from its interaction with memory. |
| **4. Procedure** | Power at matched size. Seasonal adjustment, then rerun the rules. | A test can hold size by having no power. Raw power is not comparable across oversized rules. |
| **Mechanism** | Prove exact results where the structure allows. | Simulation gives size at design points. A proof gives the limit and the reason. |
| **External** | Stock trade durations as a placebo. | This checks the account on real data. The null is unverified, so the rates are descriptive. |

# Monte Carlo design: the null is true by construction




## Why

Evidence: design of the study

- **Known null.** We set the generator, so a rejection cannot reflect a real effect.
- **Shared baseline.** All generators share one code path, so the Gaussian row is a direct baseline.
- **Stated uncertainty.** With 1,000 replications, a 5% rate has a 95% Wilson interval of about $\pm 1.4$ points. This covers counting error only.
- **Fixed in advance.** For EXP2, a design record set the hypotheses and generator checks before the main run.

# The experiments: what each one varies and what it answers

| **Name** | Question | What we vary | Size per cell |
|---|---|---|---|
| **EXP1** | Q2: calibration | $H$ from 0.5 to 0.9. Ten pairs with $H_X + H_Y$ from 1.3 to 1.7. MMD on rank-one and rank-two features. | $n \leq 4000$, 1,000–2,000 runs |
| **EXP2** | Q3: attribution | Seven generators (Gaussian, heavy-tailed, nonlinear, seasonal) crossed with $H = 0.5, 0.7, 0.8$. | $n = 500, 2000$, 1,000 runs |
| **EXP3** | Q4: power | Power with each rule tuned to a true 5%. Linear, squared and seasonal-scale links. MMD mean shift. | $n = 500, 2000$, 1,000 runs each way |
| **EXP4** | Q4: adjustment | Remove the mean, then the scale, then use an estimated period. Slow scale drift. | $n = 500, 2000$, 1,000 runs |
| **EXP5, EXP6** | Q2, Q4: robustness | Larger $n$, and a shift scaled to the null spread. Bandwidth at 0.5, 1, 2, 4 times the median. | $n \leq 8000$, 144–1,000 runs |
| **Stocks** | External check | AAPL, MSFT, NVDA trade durations, January to June 2024, windows at least 20 sessions apart. | $n \leq 2000$, 500 pairs |

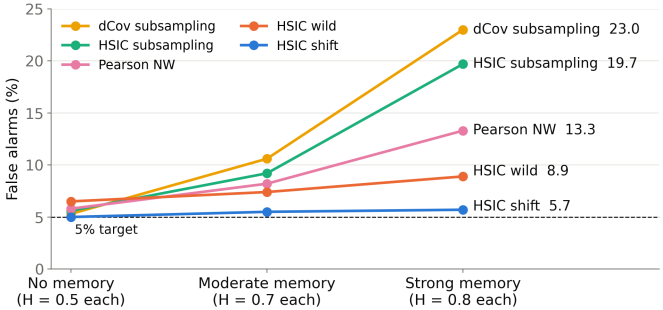
# Q1: the published conditions exclude fGn with $H > 1/2$, mostly by implication

| Source | Stated condition | fGn with $H > 1/2$ |
|---|---|---|
| Gretton et al. (HSIC, MMD), Székely et al. (dCov) | i.i.d. | Excluded |
| Chwialkowski, Gretton 2014 (shift HSIC) | $\beta$-mixing at a polynomial rate | Excluded: not strongly mixing |
| Chwialkowski et al. 2014 (wild) | $\sum_r r^2 \tau(r)^{1/2} < \infty$ | Excluded: our derivation gives $\tau(r) \gtrsim r^{2H-2}$ |
| Ghoshal, Shao 2026 (self-normalized) | Summable strong mixing | Excluded |
| Diz-Castro et al. 2026 (HSIC wild) | Near-epoch dependence on a $\beta$-mixing process | Excluded for $H_X + H_Y \geq 3/2$. Open below |
| Betken, Dehling, arXiv 2107.03041 (dCov) | Covers Gaussian-subordinated long memory | Covered in part. No Gaussian-kernel HSIC limit |

**Why** Evidence: audit of primary sources

The conditions rule these data out but give no size for the failure, so we measure it. None of the audited sources treats seasonality.

# Q2: with the same generator and $n$, the rule changes the size

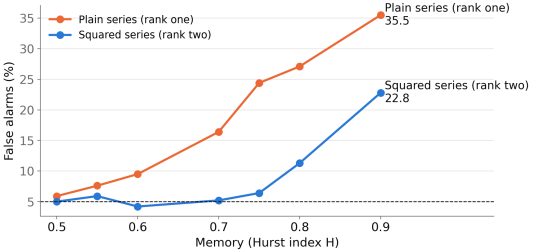



**Why** Evidence: measured, 2,000 replications per point

Only the statistic and rule differ between lines, so they account for the gaps. Part of the excess is not memory: the wild rule already sits at 6.5% at $H = 0.5$.

→ *Why do distortions start where they do?*

# Q2: higher Hermite rank delays the size distortion in split MMD

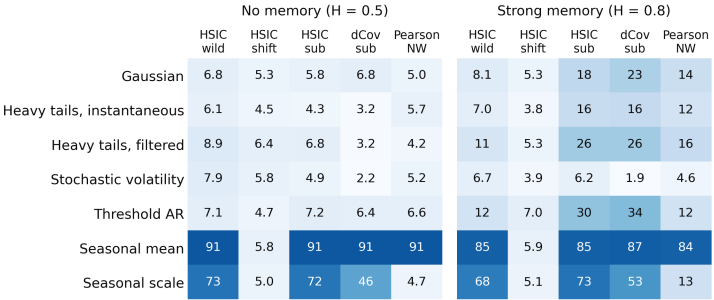



**Why** Evidence: measured, with a classical rank calculation for the shape

- Rank $m$ gives a leading covariance of order $\rho(k)^m$. Rank two balances at $H = 3/4$. Independent rank-one products balance at $H_X + H_Y = 3/2$.
- The calculation explains the shape. It gives no critical value, and the plots do not locate a proved boundary.

$\rightarrow$ *Memory distorts size. Is memory the main cause?*

# Q3: in our settings, seasonality distorts size more than memory

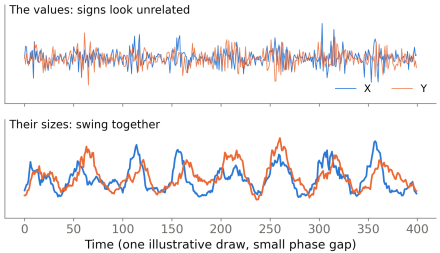


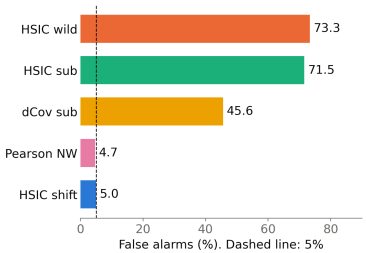

**Why** Evidence: measured, 1,000 replications per cell, Holm-corrected

We switch on one feature at a time and cross it with memory. HSIC shift stays between 3.8% and 7.0%. The threshold AR adds short-range persistence, so its excess is not purely a nonlinearity effect. The seasonal settings are illustrative and not calibrated to market data.

→ *The two seasonal forms fail for different reasons.*

# Q3: with no memory, a seasonal scale distorts kernel tests unless they use shifts



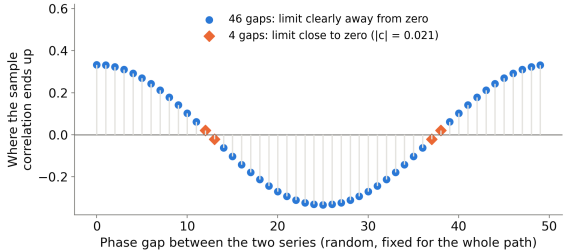




## Why

Evidence: measured, with no proof yet

These series are independent, and each has its own random phase. HSIC wild, HSIC subsampling and dCov subsampling respond to sizes and reject 46–73%. HSIC shift rejects 5.0%. Pearson sees only the linear link and rejects 4.7%.

# Q3: a seasonal mean distorts every independence rule tested except shifts

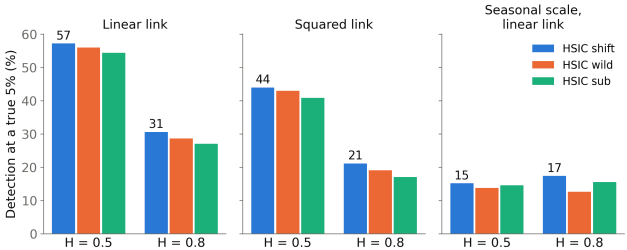



**Why** Evidence: proved for these rules (Proposition 1), measured for shifts

- One path contains one phase gap. The sample correlation converges to a random limit set by that gap.
- With $P = 50$, 46 of 50 gaps give a limit clearly away from zero, which predicts 92%. Pearson rejects 91.4%.
- Shifts re-randomize the gap and stay at 5.8%. We can only sketch why.

→ *Shifts hold size throughout. Do they do so at the cost of power?*

# Q4: shifts keep their power when every rule is tuned to a true 5%



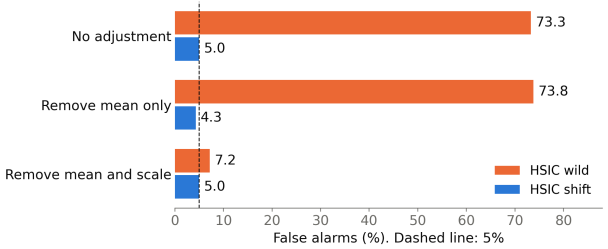

## Why

Evidence: measured at matched size

- Shifts stay within 4.9 points of the best other HSIC rule in all 18 cells.
- Raw power can mislead. HSIC subsampling shows 86.4% at its nominal critical value and 14.6% at a true 5%.
- The tuning uses the known null, so it checks the rules and is not a method for real data.

→ *Under a shared fixed seasonal phase, shifts still reject 28–34%, so we adjust first.*

# Q4: adjust first, removing the seasonal scale as well as the mean

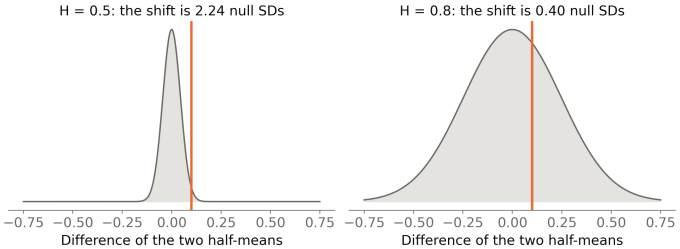



**Why** Evidence: measured, 1,000 replications per cell

- Kernel tests respond to the scale, so removing the mean alone barely changes this cell.
- A period estimated from the data worked as well as the true period.
- Adjustment costs size under memory. On Gaussian pairs at $H = 0.8$, HSIC shift goes from 5.3% to 6.8%.

→ *For split MMD, the obstacle is different: information, not calibration.*

# Q4: memory caps power against a mean shift for location-invariant tests




**Why** Evidence: proved (Proposition 2), measured for MMD

- At $n = 2000$, a shift of 0.1 is 2.24 null standard deviations at $H = 0.5$ and only 0.40 at $H = 0.8$.
- No location-invariant 5% test can pass 12% power at $H = 0.8$. MMD reaches 6.4–7.5%.
- The loss is in the data. It is not a failure of a kernel rule.

→ *The two propositions in full, and what remains unproved.*

# Formal results: what we prove and what we do not

## Proposition 1: seasonal mean, any $H \in (0,1)$

$X_t = \xi_t + a\sin\{2\pi(t+U)/P\}$, and $Y_t$ likewise with an independent phase $V$. Let $D = (U - V) \bmod P$.

- $r_n \to c(D) = \dfrac{(a^2/2)\cos(2\pi D/P)}{1 + a^2/2}$ almost surely.
- If the critical value is of smaller order than the seasonal term, the size tends to 1 at $P = 50$. This covers Pearson NW and EWC, and HSIC wild and subsampling.

**Reason:** one path fixes one gap, so the limit is random and non-zero.

## Proposition 2: half-sample mean shift

$Z_t = \mu + \xi_t + \delta\,\mathbf{1}\{t > m\}$, with $n = 2m$.

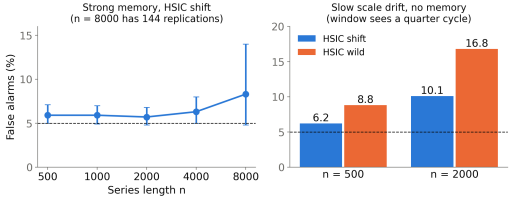
- $\mathrm{Var}(\Delta_m) = (4 - 2^{2H})\,m^{2H-2}$, exactly for every $m$.
- A level-$\alpha$ location-invariant test has power at most $1 - \Phi(z_{1-\alpha} - \delta I_n^{1/2})$. Here $I_n$ is the inverse GLS variance of $\hat{\delta}$.
- At $n = 2000$, $\delta = 0.1$: caps of 72%, 18%, 12% at $H = 0.5, 0.7, 0.8$.

**Reason:** the bound comes from the information in the data, so it holds for every such test.

## Not proved

The finite-sample size of shifts, which we only simulate. An argument shows the shift rule's size is right asymptotically (inc1, and raw below 3/2), but it rests on an unreviewed preprint. Proved directly: inc1's statistic stays bounded at every $H < 1$ at a fixed width. The seasonal-mean sketch lacks uniform convergence and ties. The RKHS Hermite expansion. The seasonal-scale failure, which we only measure.

# Q4: what is and is not proved



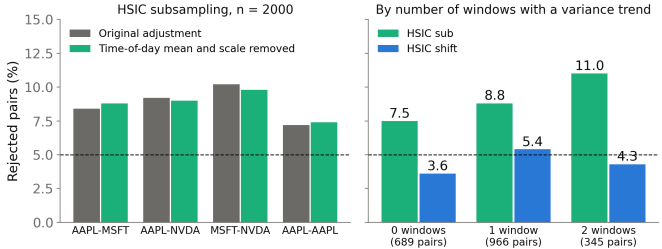

## Why

Evidence: measured, with wide intervals at large $n$

- Shift rates at $H = 0.8$ are 5.7%, 6.3% and 8.3% at $n = 2000, 4000, 8000$. The intervals overlap, and $n = 8000$ has only 144 replications.
- An argument shows the shift rule's size is right asymptotically (inc1, and raw below $3/2$), but it rests on an unreviewed preprint. Proved directly: inc1's statistic stays bounded at every $H < 1$ at a fixed width.
- A quarter-cycle window cannot be bin-adjusted, so raw has no correction. In EXP7, inc1 with shifts gives 5.8% (simulation only).

→ *Does slow drift appear in real data?*

# Stock trade durations: removing time of day barely changes the results

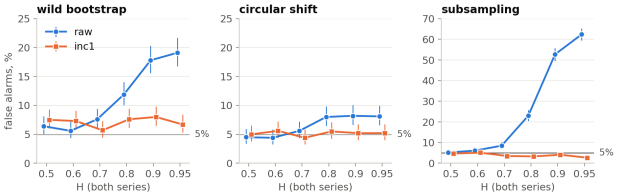



## Why

Evidence: descriptive only

- Removing the time-of-day pattern moves no cell by more than 0.6 points.
- About 44% of windows still show a variance trend, against 5% with no trend. Pairs with trends reject more under subsampling. The shift rule does not rise in step.
- We cannot verify independence and days recur, so these are not false-alarm rates.

# A kernel that forgets memory




**Why** Evidence: simulated in EXP7, 1,000 replications per point, $n = 2000$

- **Increment kernel.** It compares each point with the one before it. Changes forget the slow swings that make long memory.
- **Hermite split.** It removes the low polynomial layers of the normal scores, where memory sits. It tests those layers on their own.
- **For HSIC independence, the climb with $H$ stops.** At $H = 0.95$, wild falls from 19.1% to 6.7% and subsampling from 62.4% to 2.7%.
- The wild excess stays. Memory kernels reject 6.4–8.3% under wild, even with no memory.

→ *What it fixes and what it does not.*

# What it fixes and what it does not

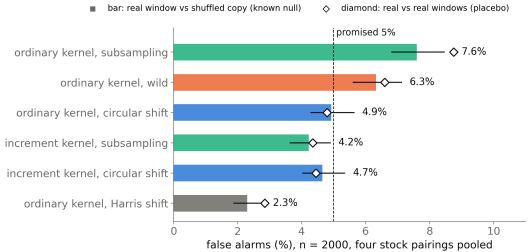
**Why** Evidence: simulated in EXP7, power at a true 5% uses the known null

- **Split MMD still climbs.** On heavy tails at $H = 0.9$, datadiff rejects 17.4%, sym 11.6%, herm1 9.4% and hsplit1 8.0%.
- **hsplit2 comes closest.** Its block rule holds on both data types. Its plug-in rule has one rising cell.
- **Each kernel has a blind spot.** inc1 misses a flat moving average (4.3% against 71.1%) and a slow common factor (8.6% against 100%). herm1 misses linear dependence (6.0% against 58.0%).
- **Harris shift is the safe baseline.** On the raw kernel it rejects 1.6–3.8% in all 44 stationary nulls. No memory kernel beats it everywhere.

**For the paper**

Present inc1 as a cure for the memory climb in HSIC, paired with shifts or Harris. It is not a fix for size.

# Real stock data with a known null (EXP8)




**Why** Evidence: EXP8, 4,000 pairs per bar, AAPL, MSFT, NVDA durations, Jan–Jun 2024

- **The design.** Pair a real window with a shuffled copy of another stock's window. The copy keeps its values and memory but loses any time alignment, so independence holds by construction.
- **Shifts hold on real data.** 4.9% with the ordinary kernel and 4.7% with inc1. Wild gives 6.3% and subsampling 7.6%.
- **The placebo matches the known null.** The real–real rates (diamonds) sit close to the bars, so most of the stock excess looks like rule error, not cross-stock links.

# EXP8: what held, what failed, what it changes

**Why** Evidence: five predictions fixed before the run, DESIGN.md amendments A1–A2

- **Held.** The windows match the saved stock windows exactly. Shifts stay in 3.4–6.8% in every $n = 2000$ cell. Harris stays at or below 3.2%.
- **Failed: memory trend.** Wild and subsampling exceed 5%, but the stock windows have mild memory ($H \approx 0.61$), too narrow a range to show a trend with $H$.
- **Failed: drift.** Keeping each copy's slow drift adds about one point, and the interval includes zero.
- **Failed: inc1 under wild.** 6.6% falls to 5.7%, not significant ($p = 0.11$). Under subsampling inc1 does fix it, 8.8% to 4.3% (not pre-registered).

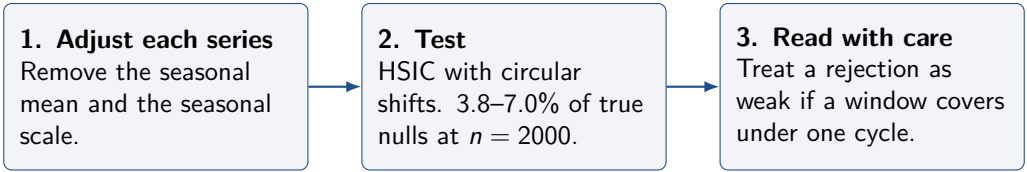
**For the paper**

The stock section can now report measured false-alarm rates, not only placebo fractions. Shifts hold at 5% on market data.

# Answers to the four research questions

| **Question** | Answer | Main evidence |
|---|---|---|
| **1. Theory** | The assumptions exclude these data by implication and give no size for the failure. | fGn with $H > 1/2$ is not strongly mixing. |
| **2. Calibration** | The rule matters as much as the statistic. | HSIC at $H = 0.8$, $n = 2000$: 19.7% subsampling, 8.9% wild, 5.7% shift. |
| **3. Attribution** | In our settings, seasonality distorts size more than memory. Heavy tails matter little. | HSIC wild at $H = 0.5$: 91% under a seasonal mean, 73% under a seasonal scale. |
| **4. Procedure** | Adjust, then use HSIC with circular shifts. Slow drift remains. | 3.8–7.0% of true nulls at $n = 2000$, within 4.9 power points at matched size. |

# The procedure for testing independence




**Why** Evidence: EXP2, EXP3 and EXP4 together

- The seasonal results give step 1, the size and power results give step 2, and the slow-drift result gives step 3.
- The 3.8–7.0% range excludes slow drift (10.1%) and a shared fixed phase (28–34%). In EXP7, inc1 with shifts gives 5.8% and 6.5% there (simulation only).
- For split MMD the rules above did not hold size under memory. The block rule of hsplit2 is one partial fix. Its plug-in rule still has one rising cell.

# What was already known, and what is new

## Already known

- The original tests assume i.i.d. data.
- Time-series extensions assume mixing, which fGn with $H > 1/2$ does not satisfy.
- The line $H_X + H_Y = 3/2$.
- Oversized dCov subsampling (Betken, Dehling).
- Seasonal-mean nonsense correlation (Yule, 1926).

## New in this paper

- How large the MMD and HSIC false alarms are, rule by rule, side by side.
- Process features crossed with memory.
- Seasonal scale as a kernel-specific failure.
- Propositions 1 and 2.
- A decision rule with power checked.

# Closest prior work: Betken and Dehling, and where we differ

| | Betken, Dehling (arXiv 2107.03041) | This paper |
|---|---|---|
| **Null, generator** | Independent long-memory processes, fGn | The same, with process features added |
| **Statistic** | Lag-weighted dCov, Pearson as comparator | Gaussian-kernel HSIC, dCov, Pearson, split MMD |
| **Critical value** | Subsampling with offset $0.1n$ | Subsampling (same offset), wild, circular shifts |
| **Theory** | dCov limit if both $H < 3/4$, or $H_X + H_Y > 3/2$ with identity transforms | No new long-memory limit. Propositions 1 and 2 are on other objects |
| **Memory, $n$** | $H = 0.6, 0.7$ and $n \leq 1000$ | $H \leq 0.9$, unequal pairs, $n = 2000$ (some to 8,000) |
| **Features** | Instantaneous transforms only | Tails, nonlinear dynamics, seasonality, crossed with $H$ |
| **dCov size** | 11.2–17.5% at $H = 0.7$ | 10.6% at $H = 0.7$, $n = 2000$ |

**Why** Evidence: audit of arXiv v3

They already have the null, the $H_X + H_Y = 3/2$ line and oversized dCov subsampling, so we do not claim these. Our distinct part is the comparison of rules, the process features, split MMD and the two propositions.

# Five questions for Wednesday's call (1 of 2): the proofs

## 1. Can we prove the size of circular shifts?

- **Where we stand.** An argument shows the shift rule's size is right asymptotically (inc1, and raw below 3/2), but it rests on an unreviewed preprint. Proved directly: inc1's statistic stays bounded at every $H < 1$ at a fixed width. The finite-sample size is simulated only. Rates at $H = 0.8$ are 5.7%, 6.3% and 8.3% at $n = 2000, 4000, 8000$. The last has 144 replications, with an interval of 4.8–14.0%.
- **Seasonal-mean sketch.** It lacks uniform convergence of $T_n(s)/n$ over offsets. It also lacks a treatment of the ties $h(d) = h(P/2 - d)$, where lower-order long-memory terms decide.
- **Ask.** Is a proof realistic, and for which case first? Or do we state the size as a measured result only?

## 2. Do the two proofs hold up?

- **Where we stand.** So far only a numerical script has checked them.
- **Steps to read first.** The ergodic lemma with periodic weights, the HSIC limit $V_n \to h(D)$ under median bandwidths, and the invariance reduction behind the power bound.
- **Citations to check.** Lehmann and Romano Theorem 6.2.1, Billingsley Theorem 2.8, and Cornfeld, Fomin and Sinai Chapter 14.
- **Ask.** Will you read the proofs appendix before we submit?

# Five questions for Wednesday's call (2 of 2): the method and the venue

## 3. Is there a correction for slow drift?

A seasonal scale with period $4n$ raises shifts to 10.1% at $n = 2000$, against 6.2% at $n = 500$, with no memory. In the stock windows, a smooth time-of-day adjustment leaves the within-window trend in place, in about 44% of windows. We have no correction for raw HSIC. In EXP7, inc1 with shifts gives 5.8% under slow drift and 6.5% under a shared fixed season (simulation only). **Ask:** do we adopt inc1 as the correction, or state slow drift as a limit?

## 4. Do we choose a data-driven bandwidth, or leave that out?

EXP6 fixes the bandwidth at 0.5, 1, 2 and 4 times the median. Wider kernels gain power under a seasonal scale and lose it against the squared link, so no fixed multiple wins. Shifts hold size at all four. We did not study data-driven choice. **Ask:** is a new experiment worth the time, or is the stated limit enough?

## 5. Which venue: a journal, or AISTATS?

The draft has 25 pages. AISTATS would mean cutting to about 8, which would lose most of the evidence. The JFec special issue date of 1 September 2026 has passed. Betken and Dehling are the closest prior work, so any venue needs our distinction from them to be clear. **Ask:** which journal would you choose, or do you prefer the short route?